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Rice Variety Classification Using Next Generation Convolutional Networks

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ABSTRACT

Rice, a staple food for over half of the global population, exhibits remarkable diversity in grain types. However, this diversity poses significant challenges in accurate identification, often resulting in misclassification and even intentional adulteration. These issues compromise product quality and erode trust within the agricultural supply chain. Traditional identification methods, which rely on manual inspection and chemical analysis, are often imprecise, labour-intensive, and impractical, especially when distinguishing grains based on external features such as size, shape, and texture. To address these limitations, we propose an advanced framework that leverages cutting-edge machine-learning techniques for rice variety classification. Our approach employs a convolutional neural network model enhanced with Transformer-based design principles, achieving exceptional performance. The model attains an accuracy of 99.60% on one dataset and 100% on another under similar experimental conditions. Its robustness and simplicity make it a highly effective solution for practical applications. The proposed methodology holds significant promise for broader applications in agriculture and food safety, paving the way for a more transparent and reliable agricultural ecosystem that fosters trust and innovation.

1 | Introduction

The integration of artificial intelligence (AI) and computer vision has revolutionized numerous industries, including agriculture [1]. Computer vision, a critical subset of AI, empowers machines to interpret and analyse visual data from digital images and videos. By leveraging advanced algorithms and deep learning models, computers can perform complex tasks such as object detection, localization, and classification with remarkable precision. This transformative capability has spurred ground-breaking innovations, particularly in the agricultural sector, where the demand for automation and precision is rapidly increasing.

Rice is one of the most essential crops globally, with unparalleled cultural, economic, and nutritional significance [2]. It serves as the staple food for more than half of the world's population, with Asia accounting for 90% of global production [3]. Countries like Bangladesh have been pivotal in advancing rice cultivation and understanding its genetic diversity [4]. Despite its importance, the rice industry faces significant challenges in efficiently identifying and classifying the thousands of rice varieties grown worldwide. Current estimates suggest that there are over 40,000 rice varieties globally [5], categorized into diverse types such as short-grain, medium-grain, long-grain, and parboiled rice. This immense diversity highlights the urgent need for effective methods to ensure variety-specific cultivation, quality control, and fraud prevention.

Traditional methods for identifying rice varieties rely heavily on manual inspection and chemical analysis, which are often labour-intensive, expensive, and time-consuming. Additionally, these methods may involve hazardous reagents, posing risks to both the environment and human health. According to the latest FAO data, the worldwide total production of rice is estimated

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to be around 519.5 million tons. This figure is based on the preliminary forecast for 2022 [6]. Consequently, the development of an automated, artificial intelligence (AI) driven solutions is not only desirable but also essential for improving efficiency, reducing costs, and ensuring environmental sustainability in rice production.

Recent advancements in computer vision have demonstrated significant potential in addressing the challenges of rice variety identification. By integrating an image acquisition, preprocessing, segmentation, and analysis, modern AI systems can not only identify and classify rice grains but also assess their quality with unprecedented precision. These systems can detect the subtle differences in shape, size, texture, and colour, enabling the classification of rice varieties that may be indistinguishable from the human eye [7]. Such innovations are particularly valuable for farmers and consumers, who often struggle to differentiate between rice varieties or assess grain quality, leaving them vulnerable to economic losses and fraudulent practices. However, the development of automated systems for rice variety identification presents its own set of challenges. The vast diversity of rice strains, along with regional variations and impurities in the grains, complicates the classification process. While research in this field is expanding, existing studies are still limited in scope and scale, highlighting the need for more comprehensive datasets, advanced algorithms, and robust models to address the complexities of rice grain identification. The pursuit of automated rice variety classification dates back to 1995, when a pioneering study [8] analysed two-dimensional images of four rice varieties, marking the beginning of ongoing innovations aimed at enhancing accuracy and reliability in this challenging domain. This marked the beginning of a series of advancements in this domain. In 2002, Wee et al. [9] introduced the fuzzy art-map neural network, achieving a commendable 95.4% accuracy in rice grain quality classification. That same year, Marini et al. [10] utilized a counter-propagation artificial neural network to predict 11 rice varieties with approximately 90% precision. Building on this, Hobson et al. [11] developed a digital image analysis technique in 2007, enabling the quantitative characterization of eight rice varieties. Further advancements came in 2008, when Guzman and Peralta [12] implemented an artificial neural network, achieving close to 97% accuracy in classifying five varietal groups in the Philippines. The use of neural network classifiers continued with Rad et al. (2011) [13], who focused on colour and texture characteristics, achieving a 96.67% accuracy rate across five rice varieties. Meanwhile, Nagendra et al. [14] explored chemical testing methods in 2013, attaining an 85% accuracy for rice classification. Huang et al. [15], in 2015, employed support vector machine (SVM) techniques to classify 79 varieties with an accuracy of 79.74%, while Aki et al. [16] achieved 90.5% accuracy using the non-nested generalization algorithm. During the same period, Sumaryanti et al. [17] demonstrated the effectiveness of the learning vector quantization algorithm, achieving 96.6% accuracy in classifying ten rice varieties. Significant progress was made in 2016, with multiple studies advancing rice classification methodologies. One study used backpropagation neural networks and wavelet decomposition, which offered an efficient classification technique [18]. Kuo et al. [19] applied the SRC classifier, yielding an accuracy of 89.1% for 30 rice varieties. Zareiforoush et al. [20] achieved an impressive 98.72% accuracy using an artificial neural network on a smaller dataset. The application of deep learning became increasingly prominent from 2017 onwards. Lin et al. [21] utilized a deep convolutional neural network (DCNN), achieving 95.5% accuracy in classifying three rice varieties. Similarly, Rexce J. et al. [22] achieved a 92.31% accuracy rate using the MLP neural network for 13 rice types. Wah et al. [23] used image segmentation combined with the K-nearest neighbour classifier to classify three varieties. In 2019, Cinar and Koklu [24] applied various AI methods, achieving 93.02% accuracy for two rice types, while Hue et al. [25] utilized morphological feature extraction, attaining 95.5% accuracy. Dheer and Singh (2019) [26] leveraged machine learning classifiers, achieving an impressive 99.16% accuracy for eight rice varieties. Moving into 2020, Ibrahim et al. [27] performed a comparative analysis of rice classification using artificial neural networks, attaining a 93.34% accuracy rate. Weng et al. [28] incorporated hyper spectral imaging and deep learning, achieving 98.57% accuracy. Recent years have seen studies leveraging advanced machine learning methods for outstanding results. For instance, Saxena et al. [29] achieved 99.85% accuracy using popular machine-learning approaches for five rice types. Concurrently, Lee and Tay [30] applied convolutional neural networks to classify seven rice varieties, while Nair et al. [31] introduced a fuzzy genetic cascading system, achieving 94% accuracy for two types of rice. Deep learning techniques gained further traction in 2022 and 2023. Jeyaraj et al. [32] proposed a deep learning framework to classify five varieties and integrated it into a real-time variety detection system. Tasci et al. [33] achieved 99.87% accuracy using the LeeNet-5-based quantized neural network, while Iqbal et al. [34] developed a lightweight MobileNetV2 model, attaining a 99.7% accuracy rate for five rice types. In 2024, Kang et al. [38] explored fluorescence hyper spectral imaging for rice classification, achieving 95.3% accuracy using a combination of backpropagation neural networks and random forest. Another study applied ensemble methods with SVM classifiers to three rice varieties, reporting 96% accuracy [37].

The datasets utilized in these studies highlight the diversity of approaches and experimental designs in rice variety classification. Initial efforts often relied on small datasets, such as the study by Wee et al. [9], which analysed a limited number of images using features like Zernike moments. Similarly, Hobson et al. [11] worked with 400 images to explore grain length and texture characteristics. In contrast, larger datasets have enabled researchers to employ more complex techniques. For instance, Weng et al. [28] used hyper spectral imaging of 4320 samples to extract spectral reflectance and morphological features, achieving remarkable results. The trend toward using expansive datasets is evident in studies like those by Tasci et al. [33] and Iqbal et al. [34], which analysed 75,000 rice grain images to classify five distinct varieties. These datasets included features such as colour, morphology, and texture, derived using advanced imaging techniques. A notable contribution by Dheer and Singh (2019) [26] involved eight rice varieties, with physical characteristics extracted from a dataset of 500 images. In terms of features, earlier datasets primarily focused on geometric properties, such as area, perimeter, and elongation [8]. However, recent datasets incorporate a broader spectrum of attributes, including spectral features, texture, and wavelet analysis, as seen in the works of Zareiforoush et al. [20] and Weng et al. [28]. Moreover, controlled imaging conditions, as adopted by Lee and Tay [30], ensured the consistency and reliability of data. Despite the growing sophistication of datasets, challenges remain. Limited geographical

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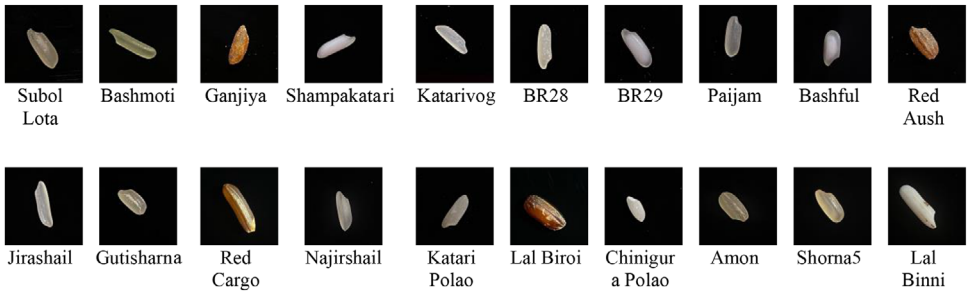


FIGURE 1 | Sample images from Dataset-01.

representation, small sample sizes, and imbalanced datasets continue to restrict the generalizability of some findings. Addressing these gaps, the current study employs a self-curated, extensive dataset, encompassing diverse rice varieties and features. This robust dataset forms the foundation for the advanced deep learning models employed, aiming to significantly enhance the classification accuracy and practical applicability of rice variety detection systems. Islam et al. [44] illustrate a comprehensive study comparison and future works in rice variety classification.

This study addresses existing gaps by harnessing the power of next-generation convolutional neural network (CNN) models and deep learning to develop an innovative rice variety classification system. A key contribution of this research is the creation of a well-structured and comprehensive dataset containing images of twenty distinct rice varieties. By leveraging optimized deep-learning models to extract diverse features from these images, the study aims to achieve exceptional classification accuracy. Notably, our applied next-generation CNN [45] represents a state-of-the-art advancement in the field of rice variety classification. The findings of this study have far-reaching implications for the agricultural industry, offering a scalable and efficient solution for rice variety identification. By transitioning from traditional methods to AI-based systems, stakeholders can achieve faster, more accurate, and environmentally friendly quality assessments. This approach not only addresses the limitations of existing techniques but also opens new avenues for research and innovation in crop management and food security.

This paper is organized as follows: The first section outlines the research introduction and a review of relevant literature. The second section presents the proposed methodology, supported by graphical illustrations, algorithmic explanations, and a dataset description. The third section provides a detailed analysis of the results, while the fourth section offers a comparative evaluation. Finally, the paper concludes with key findings and outlines directions for future research.

2 | Methodology

The proposed methodology is structured into several key stages. First, we developed a comprehensive and diverse dataset, incorporating unique visual attributes, and collected an additional well-known dataset from the public domain. Second, we implemented a next-generation CNN model, enhanced with Transformer-based attention mechanisms. The model's hyperparameters were fine-tuned, and deep features were extracted to classify rice varieties effectively.

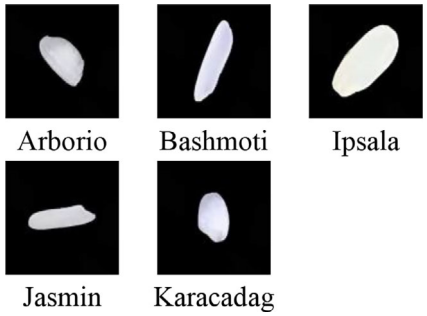


FIGURE 2 | Sample images from Dataset-02.

2.1 | Datasets Description

In this study, two distinct datasets are utilized: one is self-developed 'Dataset-01' [35, 40] and another 'Dataset-02' is the well-known Cinar and Koklu rice dataset [39]. In our self-developed dataset 'Aruzz22.5K' there are 20 rice varieties based on criteria of colour and size which are validated by agricultural experts. These varieties are known as Subol Lota, Bashmoti (Deshi), Ganjiya, Shampakatari, Sugandhi Katarivog, BR-28, BR-29, Paijam, Bashful, Lal Aush, BR-Jirashail, Gutisharna, Birui, Najirshail, Red Cargo, Polao (Katari), Polao (Chinigura), Amon, Shorna-5, and Lal Binni in local market places. Another dataset is collected from the public domain. Both dataset is split into train (80%) and test (20%) sets. A selection of sample images from our curated dataset is presented in Figure 1, and from the external dataset is shown in Figure 2. Table 1 shows the train and test split details for each dataset.

2.2 | Proposed Model

The ConvNet model architecture utilized for rice variety classification is depicted in Figure 3, is a deep-learning architecture optimized for feature extraction. Known for its efficiency with visual datasets, this model is particularly suitable for devices with limited computational power, such as mobile devices. This model combines the hierarchical design of traditional CNNs with the attention mechanisms of Transformers, enhancing its performance and versatility. The model incorporates a global average pooling (GAP) layer, which flattens the convolutional outputs into a single vector, which is used for our classification

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TABLE 1 | Datasets' descriptions.

Datasets	Total data	Classes	Train data	Test data
Dataset-01 [40]	27,000	20	22,500 (1125 images for each class)	4500 (225 images for each class)
Dataset-02 [39]	6750	5	5625 (1125 images for each class)	1125 (225 images for each class)

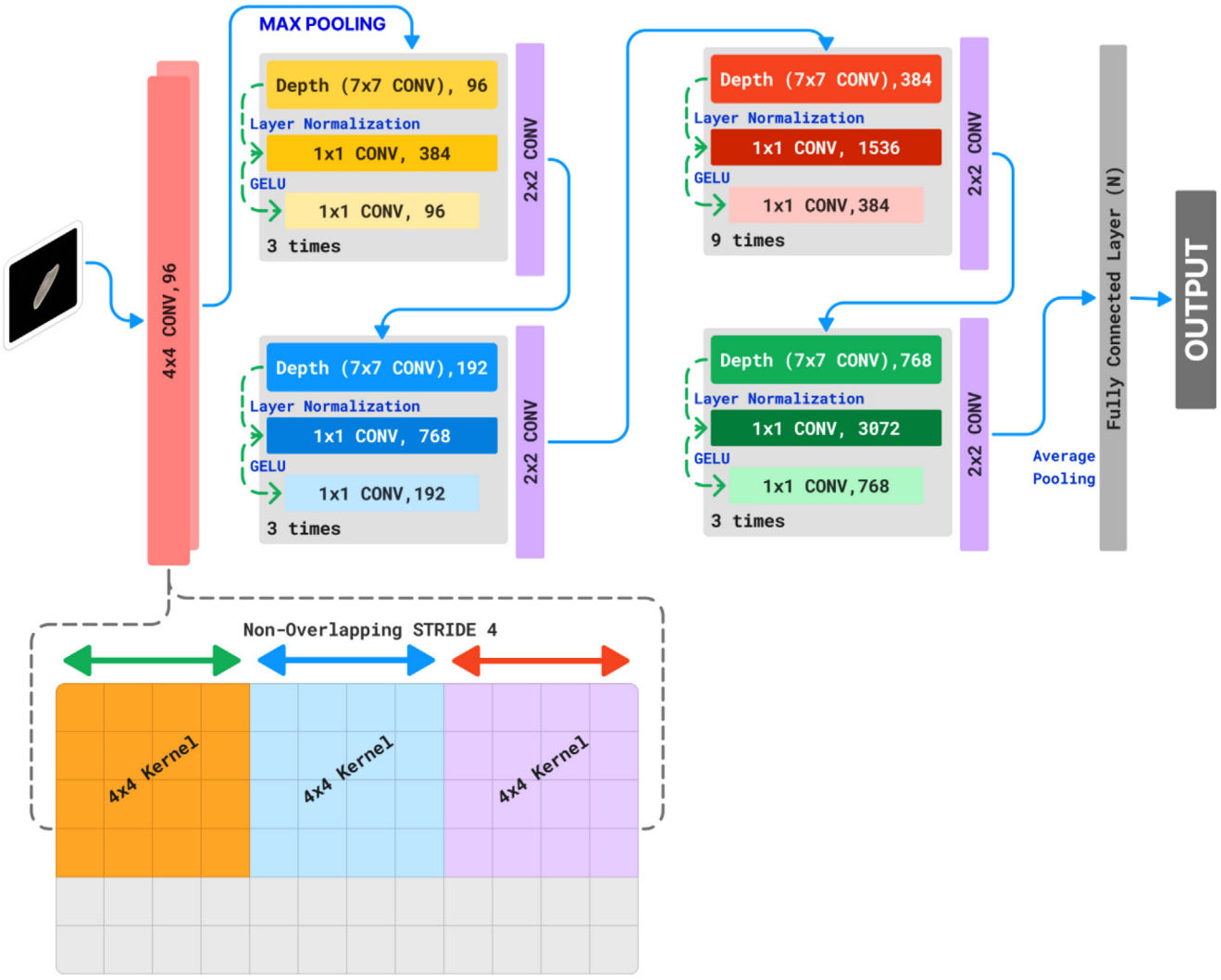


FIGURE 3 | Next generation ConvNet model architecture.

tasks by passing it through fully connected layers. By reducing the reliance on fully connected layers, the GAP layer achieves a more lightweight structure, making this model ideal for faster and resource-constrained devices. Additionally, this model integrates batch normalization and non-linear activation layers alongside convolutional and pooling layers. These enhancements improve the model's performance and robustness against overfitting. The architecture consists of stacked convolutional layers, normalization, and nonlinear activation functions. The applied model employs depth-wise separable convolutions, layer normalization (LN), inverted bottleneck structures, and large kernel sizes for improved performance.

First, a standard convolutional operation is done on a given image dataset, which can be represented by Equation (1)

$$\left[y\ [i,j,k] = \sum_{c=1}^{C} \sum_{m=-M}^{M} \sum_{n=-N}^{N} w\,[m,n,c,k]\ \cdot x\,[i+m, j+n, c] \right], \quad (1)$$

where,

$(x[i,j,c])$: Input tensor at the spatial position $((i,j))$ and channel (c).
$(w[m,n,c,k])$: Convolutional kernel of size $((2M+1)\times(2N+1))$.
$(y[i,j,k])$: Output tensor at position $((i,j))$ and channel (k).
M, N: Kernel size along with height and width.

The proposed model employs a kernel size of 4×4 with a stride of 4, a configuration that enhances performance on complex visual

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TABLE 2 | Classification report of Dataset-01.

Class name	Precision	Recall	*F*1-score
Subol_Lota	1.00	1.00	1.00
Bashmoti	1.00	1.00	1.00
Ganjiya	1.00	0.99	0.99
Shampakatari	1.00	1.00	1.00
Katarivog	1.00	1.00	1.00
BR28	1.00	0.99	1.00
BR29	0.98	1.00	0.99
Paijam	0.99	1.00	0.99
Bashful	1.00	1.00	1.00
Lal_Aush	1.00	1.00	1.00
Jirashail	1.00	1.00	1.00
Gutisharna	1.00	0.99	1.00
Red_Cargo	1.00	1.00	1.00
Najirshail	1.00	0.97	0.99
Katari_Polao	0.99	1.00	0.99
Lal_Biroi	1.00	1.00	1.00
Chinigura_Polao	1.00	0.99	1.00
Amon	0.99	1.00	1.00
Shorna5	1.00	1.00	1.00
Lal_Binni	1.00	1.00	1.00

TABLE 3 | Classification report of Dataset-02.

Class name	Precision	Recall	F1-score
Arborio	1.00	1.00	1.00
Basmati	1.00	1.00	1.00
Ipsala	1.00	1.00	1.00
Jasmine	1.00	1.00	1.00
Karacadag	1.00	1.00	1.00

tasks. By setting the kernel size and stride to identical values, the model ensures non-overlapping convolutions. Consequently, as the kernel moves across the input, no information is shared between adjacent patches, resulting in the generation of distinct, non-redundant regions of the feature map. This approach facilitates the extraction of unique patterns, contributing to improved model performance in handling intricate visual data. A depth-wise convolution (Equation (2)) is applied to each input channel because it reduces computational complexity and a 1×1 point-wise convolution (Equation (3)) is applied to combine information across channels. Here,

$$\left[y_{\text{depth}}\ [i,j,c] = \sum_{m=-M}^{M} \sum_{n=-N}^{N} w_{\text{depth}}\,[m,n,c]\ \cdot x\,[i+m, j+n, c] \right], \tag{2}$$

$$\left[y_{\text{point}}\ [i,j,k] = \sum_{c=1}^{C} w_{\text{point}}\,[c,k]\ \cdot y_{\text{depth}}\,[i,j,c]. \right], \tag{3}$$

where,

$[y_{\text{depth}}[i,j,c]$: Output of depth-wise convolution at position (i,j) and channel (c).
$w_{\text{depth}}[m,n,c]$: Depthwise convolution kernel for each input channel.
$[y_{\text{point}}[i,j,k]$: Output after pointwise convolution at position (i,j) and channel (k).
$w_{\text{point}}[c,k]$: Pointwise convolution weights for transforming depth-wise features.

Instead of using batch normalization in the ConvNet, a LN is used to improve consistency across input dataset sizes. Equation (4) presents the LN technique.

$$\left[\widehat{x_i} = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}}, \quad \mu = \frac{1}{C}\sum_{i=1}^{C} x_i, \quad \sigma^2 = \frac{1}{C}\sum_{i=1}^{C} (x_i - \mu)^2 \right], \tag{4}$$

where,

$\widehat{x_i}$: Normalized activation.
x_i: Input feature.
μ: Mean of activations across the feature dimension.
σ_2: Variance of activations.
C: Number of features.
ϵ: Small constant for numerical stability.

After that, a non-linear activation function, Gaussian error linear unit (GELU), which can be represented by Equation (5),

$$(\text{GELU}\,(x) = x\ \cdot \Phi\,(x))\,, \tag{5}$$

where, $\Phi(x)$ is the cumulative distribution function of a Gaussian distribution.

The activation function for non-linearity can be represented by Equation (6).

$$\text{GELU}\ (x) = 0.5x\left(1 + \tanh\left(\sqrt{\frac{2}{\pi}}\left(x + 0.044715x^3\right)\right)\right) \tag{6}$$

In our applied model, an inverted bottleneck structure is introduced, which expands the number of channels before applying depthwise convolutions, and then reduces the number of channels afterward. If (C) is the input channels, (tC) represents expanded channels, where $(t > 1)$ is the expansion factor. The sequence of operations is:

i. Expand channels: $(\text{Conv1} \times 1(C \to tC))$,
ii. Apply depthwise convolution: $(\text{ConvKxK}(tC))$,
iii. Reduce channels: $(\text{Conv1} \times 1(tC \to C))$.

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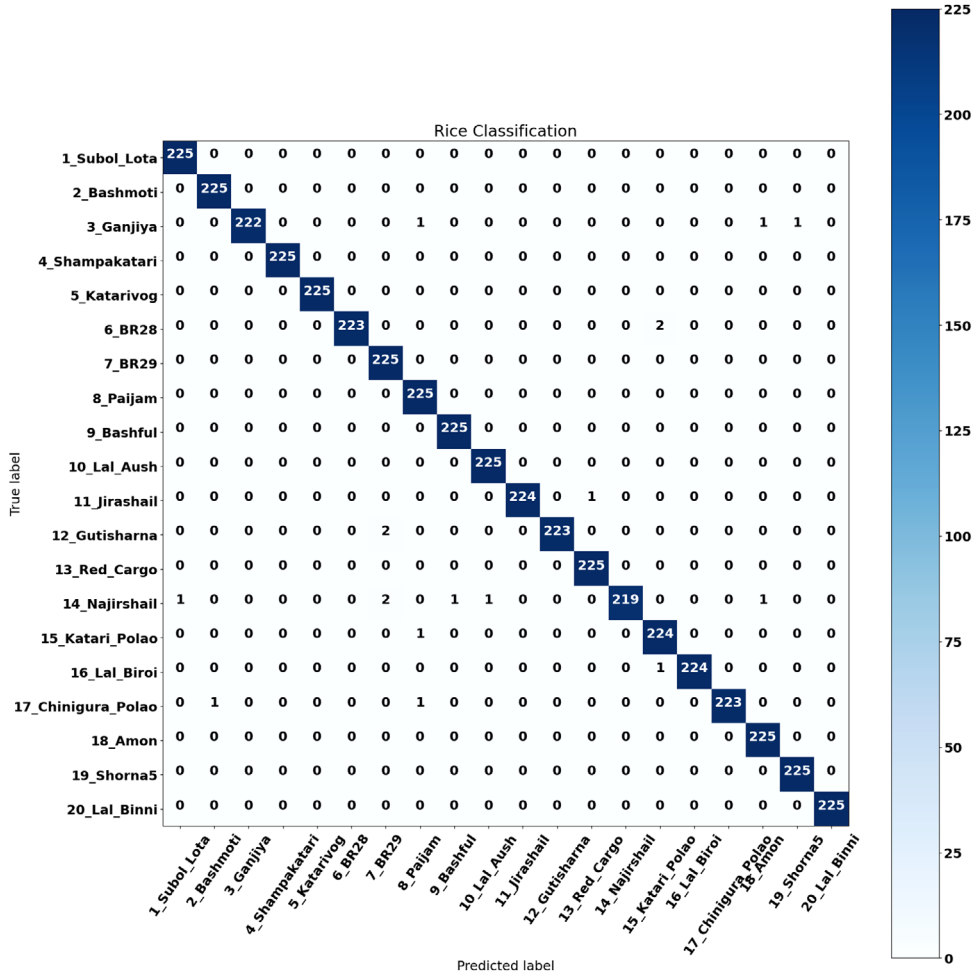


FIGURE 4 | Confusion matrix for Dataset-01.

For our multi-class classification task, we have used cross-entropy loss function shown in Equation (7).

$$\left[\mathcal{L} = -\sum_{i=1}^{N}\sum_{k=1}^{C} t_{i,k}\log\left(p_{i,k}\right)\right], \tag{7}$$

where, $(t_{i,k})$ means one-hot encoded ground truth for class k of sample I, and $(p_{i,k})$ means predicted probability for class k of sample i.

2.3 | Hyper-Parameter Settings

To optimize performance, we systematically fine-tuned key parameters:

i. Epochs (E): Preliminary tests showed peak accuracy at epoch 22, leading us to extend training to 25 epochs for refinement.

ii. Batch size (B): Experiments with batch sizes $B \in \{16, 32, 64, 128\}$ identified $B = 32$ as the optimal choice, balancing computational efficiency and model precision.

iii. Learning rate (η): A learning rate of $\eta = 0.001$ caused instability, while values below $\eta = 0.0001$ slowed convergence without significant performance gains. The optimal value was determined to be $\eta = 0.0001$, ensuring stable and efficient training.

We utilized the Adam optimizer (Equation (7)), which computes adaptive learning rates (η_t) based on the first (m_t), second (v_t) moments of the gradients, and β is the decay rate for the first moment.

$$\eta_t = \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon}, \quad \text{where} \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \tag{7}$$

2.4 | Performance Metrics

To assess the performance of our applied ConvNet, we utilized a confusion matrix that provided four key metrics: precision, recall (sensitivity), $F1$-score, and accuracy. These metrics evaluate the model's classification effectiveness for distinguishing rice grain varieties. Precision (P) measures the proportion of correctly identified instances for a specific variety. Recall (R), known as sensitivity, it quantifies the model's ability to capture all actual instances of a variety. $F1$-score provides a harmonic mean of precision and recall, balancing both metrics. Accuracy (A) reflects the overall correctness of the model's predictions. All these

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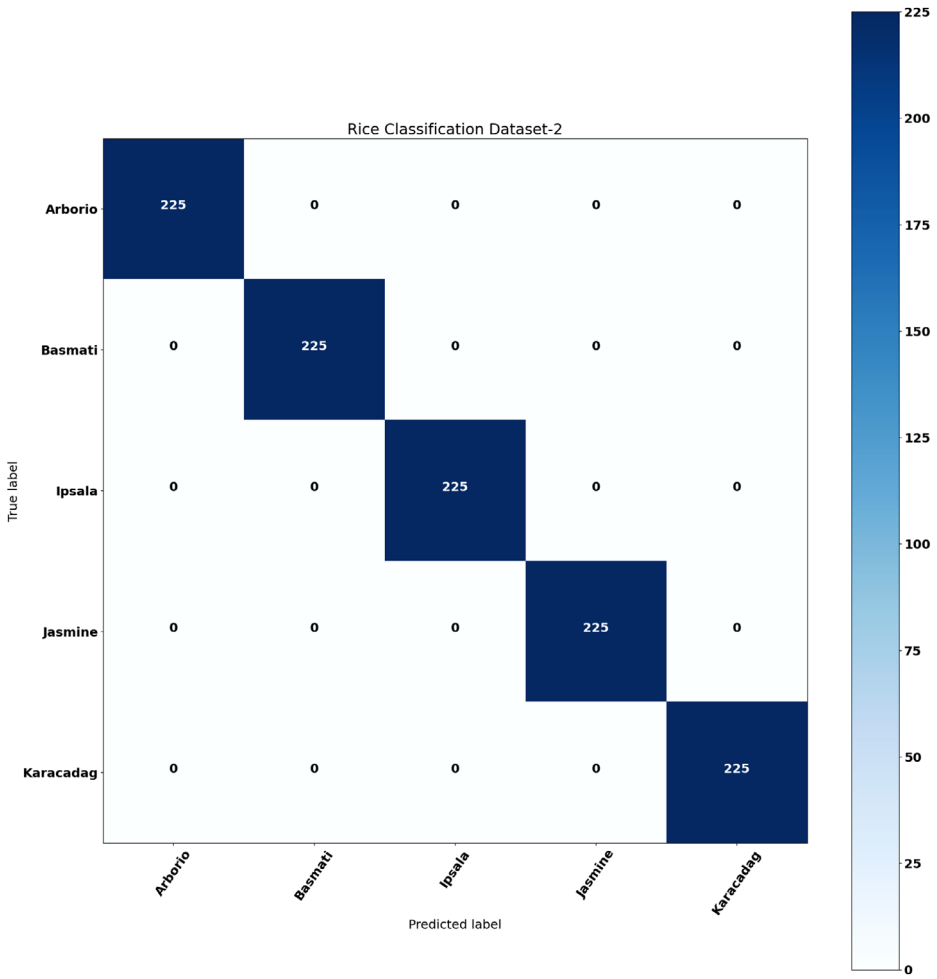


FIGURE 5 | Confusion matrix for Dataset-02.

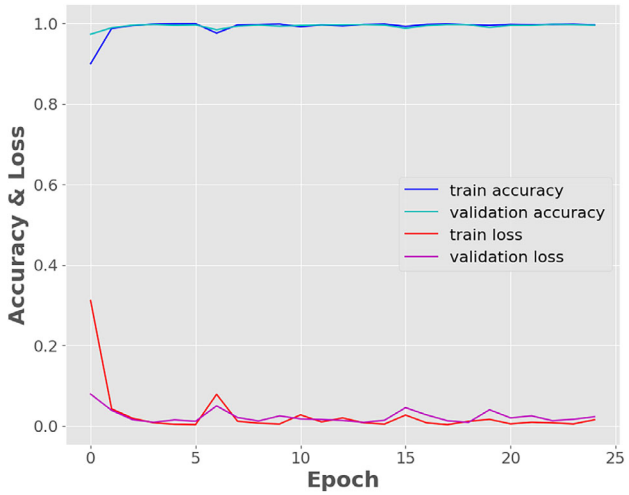


FIGURE 6 | Training and test, accuracy and loss graph for Dataset-01.

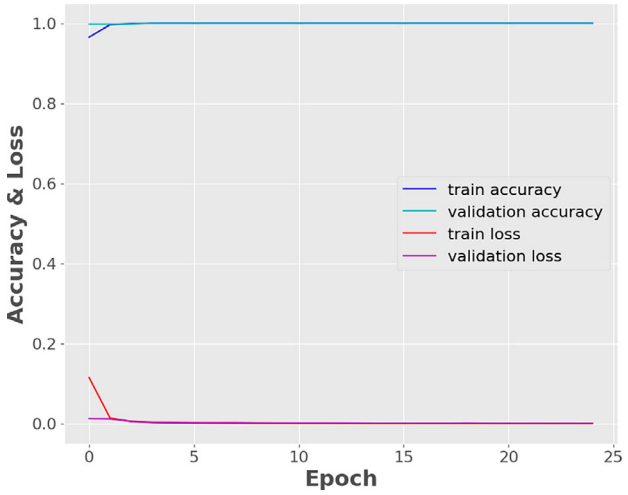


FIGURE 7 | Training and test, accuracy and loss graph for Dataset-02.

metrics can be represented by the following equations (8)–(11) consecutively.

$$P = \frac{\text{True positives (TP)}}{\text{True positives (TP)} + \text{False positives (FP)}} \quad (8)$$

$$R = \frac{\text{True positives (TP)}}{\text{True positives (TP)} + \text{False negatives (FN)}} \quad (9)$$

$$F1 = 2 \times \frac{P \times R}{P + R} \quad (10)$$

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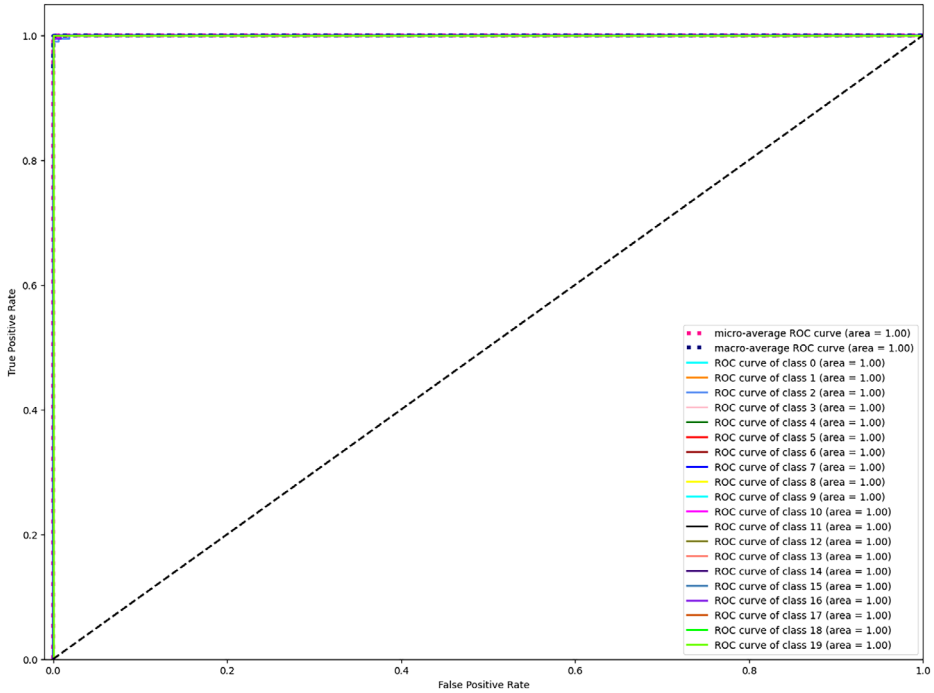


FIGURE 8 | Receiver operating characteristic curves for Dataset-01.

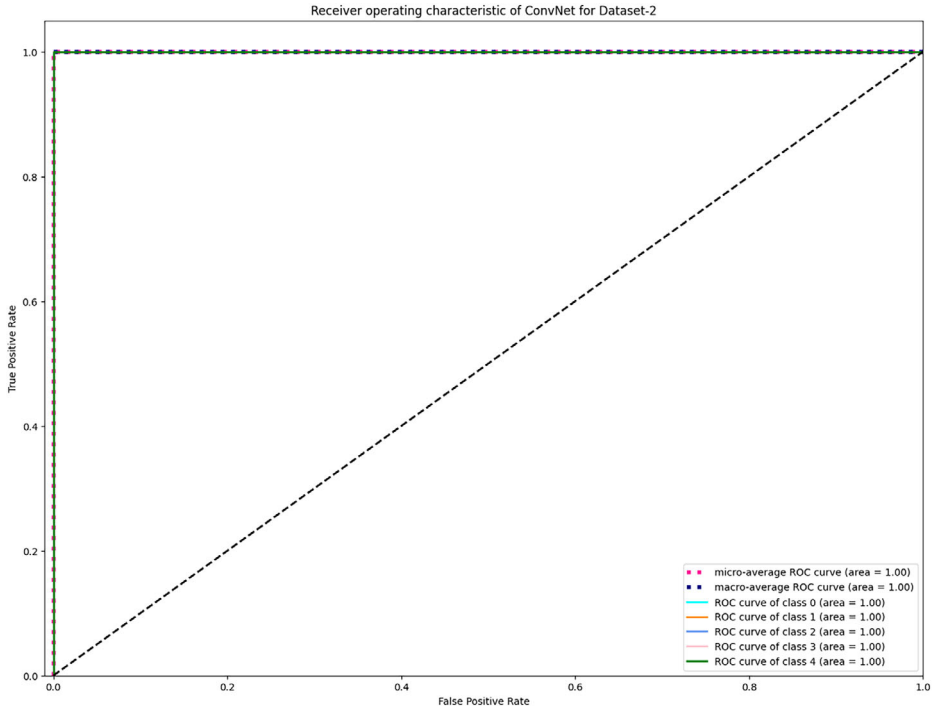


FIGURE 9 | Receiver operating characteristic curves for Dataset-02.

$$A = \frac{\text{True positives (TP)} + \text{True negatives (TN)}}{\text{Total instances } (N)} \tag{11}$$

3 | Result and Discussion

This section presents the evaluation methods and discusses the experimental results. Our ConvNet model was evaluated on two datasets using consistent training parameters, including the Adam optimizer, categorical cross-entropy (CCE) loss function, a batch size of 32, and a learning rate of 0.0001 over 25 epochs. For Dataset-01, which comprises 20 rice varieties, the model achieved a test accuracy of 99.6% with a test loss of 0.0232, demonstrating its ability to handle complex and diverse datasets. The training process took 7170.4 s (approximately 2 h) on the P100 GPU, reflecting the dataset's larger size and its complexity. In contrast, Dataset-02, consisting of five rice varieties, achieved a perfect test accuracy

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TABLE 4 | Comparative result analysis.

Author	Data size	Varieties	Classification methods	Accuracy
Marini et al. [10]	1779	11	Counter-propagation artificial neural network (CP-ANN)	91%–99%
Mousavi et al. [41]	500	5	Segmentation and backpropagation neural network-based classifier	96.67%
Mousavi et al. [42]	1500	5	Segmentation and backpropagation neural network-based classifier	98.40%
Qadri et al. [43]	10,800	6	Logistic model tree, regression, J48 tree, meta bagging, meta attribute selected classifier	97.4%
Lin et al. [21]	7399	3	DCNN	95.50%
Cinar et al. [24]	3810	2	LR, MLP, SVM, DT, RF, NB and k-NN	93.02%
Ibrahim et al. [27]	90	3	SVM and ANN	93.34%
Weng et al. [28]	4320	10	Principal component analysis network	98.57%
Nair et al. [31]	3810	2	Genetic fuzzy cascading system	94.36%
Jeyaraj et al. [32]	857	9	CNN	96.40%
Saxena et al. [29]	75,000	5	Random forest	99.85%
Iqbal et al. [34]	75,000	5	CNN	99.20%
Tasci et al. [33]	75,000	5	Quantized neural network	99.87%
Setiawan et al. [37]	45,000	3	Ensemble learning	96.74%
Kang et al. [38]	550	5	Principal component analysis, BP neural network, random forest	95.30%
	27,000	**20**	**Proposed ConvNet method**	**99.60%**
	6750	**5**		**100%**

of 100% with an exceptionally low test loss of 8.94×10^{-4}. This flawless performance can be attributed to the dataset's smaller size and greater homogeneity. The training runtime for Dataset-02 was significantly shorter, requiring only 1826 s (approximately 30 min) on the T4 × 2 GPU. These results highlight the model's robustness, scalability, and efficiency, showcasing its ability to deliver high accuracy and minimal error across varying dataset complexities. Tables 2 and 3 represent the classification reports for Dataset-01 and Dataset-02, respectively that further validate the model's exceptional performance. In the case of Dataset-01, precision, recall, and *F*1-scores across all twenty classes are nearly perfect, ranging from 0.97 to 1.00. Notably, challenging classes such as 'Najirshail' and 'Shampakatari' exhibit slight deviations but remain within the high accuracy bounds. For Dataset-02, all five classes achieve perfect precision, recall, and *F*1-scores of 1.0. The compactness of Dataset-02, combined with well-defined class features, likely contributes to this flawless classification.

The confusion matrices for both Dataset-01, shown in Figure 4, and Dataset-02, shown in Figure 5, provide critical insights into model performance across all classes. For Dataset-01, the confusion matrix shows that the model accurately predicted the majority of the samples in their respective classes, with only a few instances of minor misclassifications observed in specific challenging classes, such as 'Najirshail' and 'Ganjiya.' These misclassifications could be attributed to the slight similarities in visual features between these rice varieties. Nonetheless, the overall distribution of predictions aligns closely with the true labels, reflecting the model's ability to effectively differentiate between 20 diverse rice varieties. For Dataset-02, the confusion matrix demonstrates perfect classification, with all samples correctly identified in their respective classes. This indicates the model's exceptional capacity to distinguish between the five rice types, likely due to their distinct and less complex features compared to Dataset-01. The absence of misclassifications in Dataset-02 underscores the model's potential for achieving perfect results when dealing with smaller, more homogenous datasets.

The accuracy and loss curves for both datasets, shown in Figures 6 and 7 reflect the training process's convergence. For Dataset-01, the accuracy curve stabilizes at 99.6% with minimal fluctuations, indicating that the model has learned the underlying data distribution effectively without over-fitting. The loss curve corroborates this, showing a steady decline to near-zero values. Dataset-02 exhibit similar trends, with the model achieving 100% accuracy and negligible loss during the training and validation phases. The ROC curves for both datasets illustrated the model's discriminatory power, shown in Figures 8 and 9.

4 | Result Comparison

Table 4 provides a comparative analysis of rice variety classification studies conducted over the past 2 decades. Many of

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these studies relied on either self-constructed datasets or widely used public datasets. The comparison highlights that while some studies included a significant number of rice varieties, their dataset sizes were relatively small. Conversely, other studies utilized larger datasets but lacked diversity in the included rice varieties. Overall, most research efforts struggled to maintain a balanced combination of dataset size, variety, and diversity in morphological and colour-textural characteristics.

5 | Conclusions

This article presents an advanced approach to rice variety classification using a next-generation convolutional neural network (ConvNet) enhanced with Transformer-based principles, representing a state-of-the-art solution in this domain. By employing an advanced methodology and optimizing hyperparameters, the model achieved remarkable accuracies of 99.6% and 100% across two datasets, demonstrating its robustness and versatility. This study bridges the gap between artificial intelligence and agriculture, addressing traditional limitations in rice classification, such as labour-intensive methods and imprecision, and promoting a shift toward automated, efficient, and sustainable practices. Future research could focus on expanding datasets, enhancing model scalability, and exploring advanced architectures like Vision Transformers. This study establishes a robust foundation for precision agriculture, providing a scalable solution for quality assurance and supply chain transparency in rice classification.

Author Contributions

Md. Masudul Islam: writing - review and editing, writing - original draft, visualization, software, methodology, investigation, formal analysis, data curation, conceptualization. **Galib Muhammad Shahriar Himel:** writing - review and editing, validation, software, resources, methodology, data curation. **Golam Moazzam:** supervision, project administration. **Mohammad Shorif Uddin:** Writing - review & editing, validation, supervision, project administration, conceptualization

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data openly available in a public repository that issues datasets with DOIs

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